

Cleaning steps

The purpose of this exercise is to use LibreOffice Calc to clean messy data.

The dataset is from <https://www.kaggle.com/davidfuenteherraz/messy-imdb-dataset/data>.

	A	B	C	D	E	F
1	IMBD title ID	Original title	Release year	Genre	Duration	Country
2	tt0111161	The Shawshank Redemption		1995-02-10 Drama		142 USA
3	tt0068646	The Godfather	09-21-1972	Crime, Drama		175 USA
4	tt0468569	The Dark Knight	23-07-2008	Action, Crime, Drama		152 US
5	tt0071562	The Godfather: Part II		1975-09-25 Crime, Drama		220 USA
6	tt0110912	Pulp Fiction		1994-10-28 Crime, Drama		USA
7	tt0167260	The Lord of the Rings: The Return of the King	22 Feb 04	Action, Adventure, Drama		201 New Zealand
8	tt0108052	Schindler's List		1994-03-11 Biography, Drama, History	Nan	USA
9	tt0050083	12 Angry Men		1957-09-04 Crime, Drama		96 USA
10	tt1375666	Inception		2010-09-24 Action, Adventure, Sci-Fi		148 USA
11	tt0137523	Fight Club	10-29-99	Drama	Inf	UK
12	tt0109830	Forrest Gump		1994-10-06 Drama, Romance		142 USA
13	tt0120737	The Lord of the Rings: The Fellowship of the Ring		2002-01-18 Action, Adventure, Drama	178c	New Zealand

Fig. 1, first glance at the dataset

A preliminary review revealed several data quality issues, including misspelled country names, inconsistent date formats, missing or corrupted values, typos in category labels, and mixed data types within numeric columns.

Handling missing values and data impurities

D	E
Genre	Duration
3 Crime, Drama	
1 Biography, Drama, History	Nan
Drama	Inf
3 Action, Adventure, Drama	178c
7 Action, Sci-Fi	NULL
9 Action, Adventure, Fantasy	Not Applicable
Drama	-

Fig. 2, errors in the duration column

The duration column contained severe formatting errors, ranging from double quotes to alphanumeric strings, such as numbers mixed with letters. For a data analyst, there are generally two options: research true values or convert corrupted cells to NULL or empty. Given the small scale of this dataset, researching and manually imputing the true movie runtimes was selected as the best approach to maximize data completeness. If the dataset has thousands of rows, then automated imputation by setting the rows to NULL would have been the solution.

F	G
Country	Content Rating
Italy1	#N/A
South Korea	#N/A
France	#N/A
USA	#N/A
France	#N/A
Japan	#N/A
Italy	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
India	#N/A
USA	#N/A
USA	#N/A
UK	#N/A
UK	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
USA	#N/A
Italy	#N/A
USA	#N/A

Fig. 3, errors in content rating column

Missing entries and manually inputted values like "#N/A" were identified in Content Rating column. To maintain consistency across categorical groupings, these entries were imputed as "Unrated" so they align with other unclassified movies in the dataset.

Column standardization

I	J	K	L	M
Income	Votes	Score		
\$ 28815245	2.278.845	9.3		28815245
\$ 246120974	1.572.674	9.2		246120974
\$ 1005455211	2.241.615	9		1005455211
\$ 408,035,783	1.098.714	9.0		
\$ 222831817	1.780.147	8.9		222831817

Fig. 4, error in the Income column

Numeric columns were evaluated for non-numeric artifacts such as currency symbols (\$), commas, spaces, and accidental letters. A notable error occurred in the record for "The Godfather: Part II", where a letter "o" was mistyped in place of a digit, alongside extra commas. To test and isolate numeric values, a validation formula was used in a helper column: `=IFERROR(VALUE(TRIM(SUBSTITUTE(SUBSTITUTE(I5, "$", ""), ", ", ""))), "")`.

This formula stripped currency symbols, removed commas, trimmed whitespaces, and converted valid strings into raw numbers. The typo was corrected manually after verifying the movie's revenue.

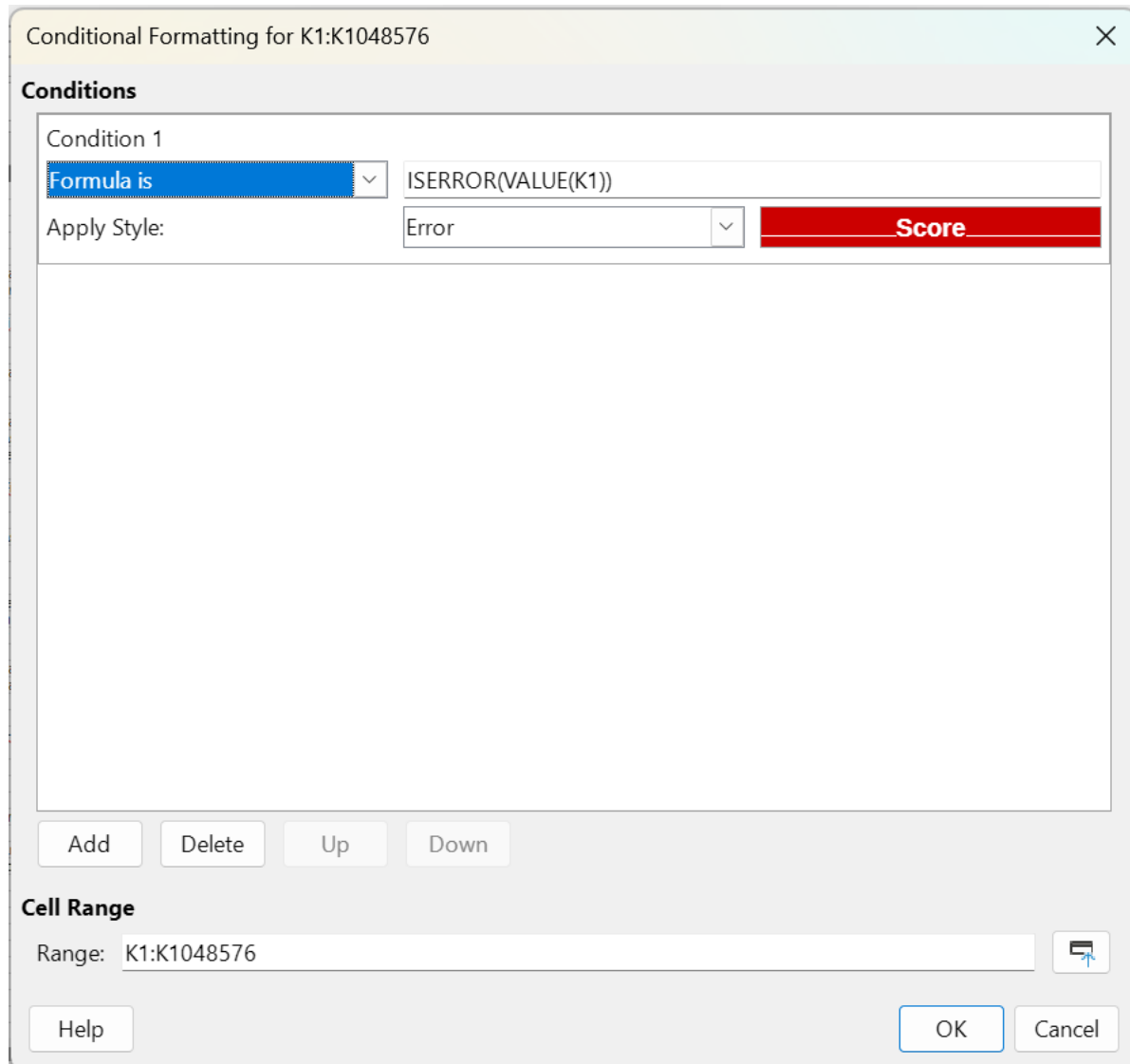


Fig. 5, conditional formatting using Formula is

	J	K
	Votes ▼	Score ▼
	2.278.845	9.3
	1.572.674	9.2
	2.241.615	9.0
	1.098.714	9,.0
	1.780.147	8,9f
	1.604.280	8.9
	1.183.248	8.9
	668.473	8.9
	2.002.816	8..8
	1.807.440	8.8
	1.755.490	8:8
	1.619.920	8.8
	672.499	8.8
	1.632.315	++8.7
	1.449.778	8.7.
	1.132.073	8,7e-0
	991.505	8.7
	891.071	8.7
	1.449.256	8.6
	1.402.015	8.6
	1.234.134	8,6
	1.204.107	8.6

Fig. 6, records with errors identified via conditional formatting

The Score column contained multiple dots, symbols, and irregular text formats. To highlight invalid entries, Conditional Formatting was applied using the formula **=ISERROR(VALUE(K2))**. This flagged every non-numeric value in red. Because the number of corrupted rows was minimal, values were manually standardized.

Spelling inconsistencies were identified across country names. Because the intended country names were obvious (e.g., minor typos like "New Zeland"), Find & Replace was used to standardize all entries to consistent, official country names.

Date standardization

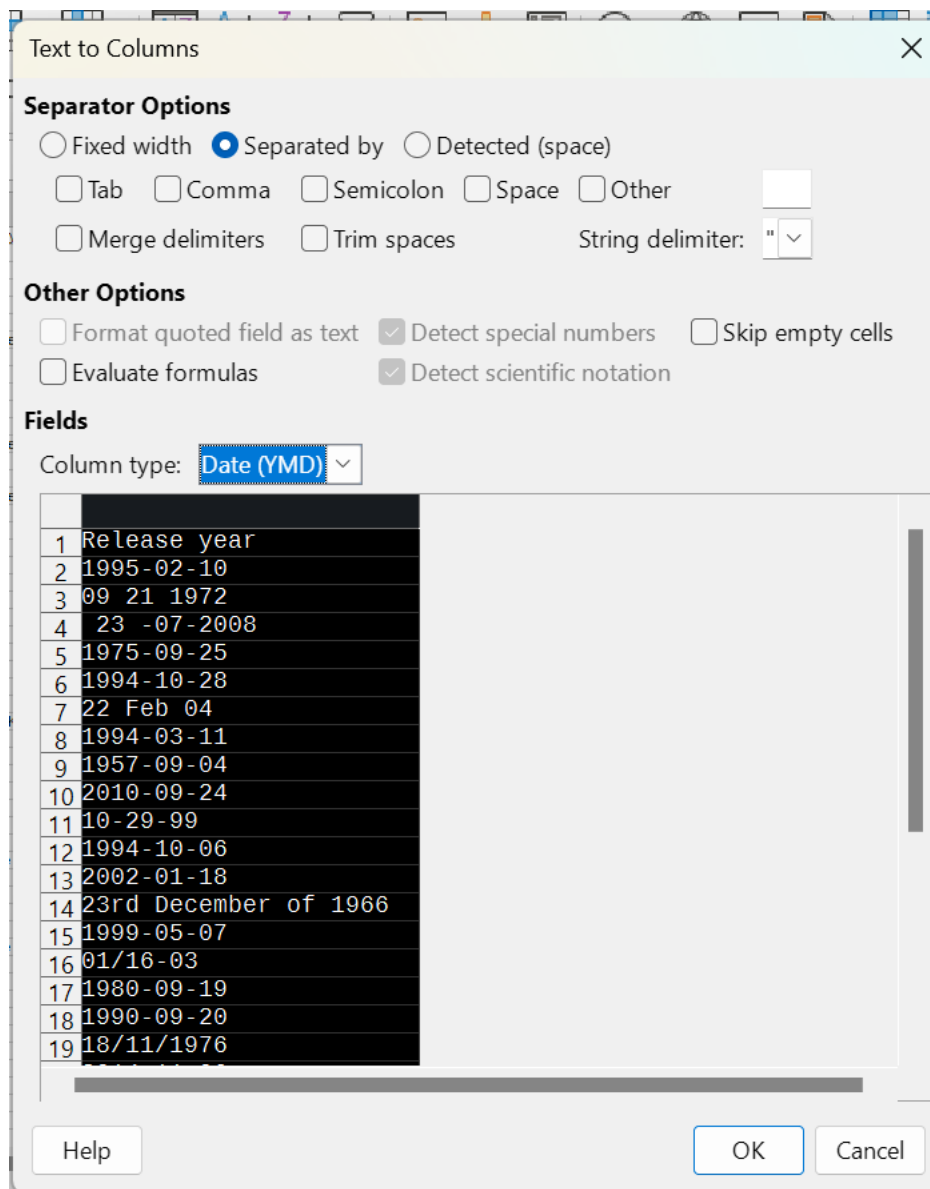


Fig. 7, the column has been selected and then picked the option of Text to Columns from the Data menu

The date column was imported as raw text and exhibited inconsistent formats (e.g., 23rd December 1947, 10-29-99, and so on).

1. The **Text to Columns** tool (with delimiters disabled) was used to make Calc parse string dates into internal date serial values.
2. For dates that failed to parse due to ordinal suffixes (st, nd, rd, th), leading spaces, or mixed slashes (/) and hyphens (-), **Find & Replace** (Ctrl+H) was used to strip the text suffixes, remove spaces, and convert slashes to hyphens.

3. Once all cells were successfully recognized as true date numbers, standard cell formatting was applied to enforce the YYYY-MM-DD standard across the entire column.